Extended reading — Week 12: Ranking and Two-Sided Matching

October 6, 2026

Five papers, ranked by deploy-relevance

1. Joachims, Swaminathan, Schnabel — *Unbiased Learning-to-Rank with Biased Feedback* (WSDM 2017, arXiv:1608.04468)

The paper that put counterfactual learning-to-rank on the production map. Introduces the position-bias model, the IPS-LTR estimator, and the dual EM-style estimation of the propensity vector p_k.

Why it matters this week. The IPS-LTR estimator in our `project/cltr.py` is the WSDM 2017 estimator. The dual learning extension (Ai et al. SIGIR 2018) builds directly on this paper; the 2024 DR-CLTR (Gupta, Oosterhuis, de Rijke WSDM 2024) extension uses the same position-bias model with a doubly-robust correction.

> "We show that the bias in the click data can be eliminated if we know the propensities p_k of observing relevance feedback. The propensities can be estimated either by intervention experiments (RandTopN) or by joint EM estimation alongside the ranker."

Commentary. The RandTopN intervention is the gold standard for p_k estimation; deploy it on a small fraction of traffic (typically 1–5%) and you get unbiased propensity estimates as a byproduct. The EM joint estimation is cheaper but assumes the position-bias model is correctly specified — which it often is not (cascade effects, multi-tab viewing). Trust direct intervention.

2. Ie, Jain, Wang, Narvekar, Agarwal, Wu, Cheng, Chandra, Boutilier — *SlateQ: A Tractable Decomposition for Reinforcement Learning with Recommendation Sets* (IJCAI 2019)

The Google paper that made slate-RL practically tractable. The key trick: under a single-choice user model, $Q(s, \mathbf{A}) = \sum_a P(a \mid \mathbf{A}, s) Q(s, a)$. The slate-action Q-function decomposes into a sum of per-item Q-functions, and the slate optimisation reduces to a fractional knapsack.

Why it matters. For the ATS slate (top-50 candidate list), SlateQ is the right framework if you want to optimise long-run reward (hire + retention) rather than instantaneous click probabilities. The per-item Q-functions can be learned with standard DQN / actor-critic techniques.

> "The single-choice user model is a strong assumption, but it allows the slate action-value function to be decomposed into a weighted sum of per-item action-value functions, where the weights are determined by the user choice model."

Commentary. The single-choice assumption fails when users consider multiple items per slate (recruiters do, every time). The Mehrotra et al. 2020 extension relaxes it via a multi-choice user model; the empirical gap is usually small ($\sim$ 1–3% NDCG) and the single-choice version is the right place to start.

3. Chen, Beutel, Covington, Jain, Belletti, Chi — *Top-K Off-Policy Correction for a REINFORCE Recommender System* (WSDM 2019)

YouTube's live-deployed RL-recsys paper. Introduces the $\lambda_K = K(1-\pi(a|s))^{K-1}$ correction for slate-of-K policy gradients, plus importance-ratio clipping and a behaviour-policy regulariser.

Why it matters. The top-K correction is the single most important piece of off-policy machinery for slate recommenders. Without it, the policy gradient overweights items that the new policy already favours (because they would still be in the top-K even without training push), and underweights tail items (where the marginal training gain actually lives).

> "We find that the top-K correction $\lambda_K(a|s) = K(1-\pi(a|s))^{K-1}$ substantially improves stability and final performance over vanilla off-policy REINFORCE."

Commentary. Two things to note. First, the correction assumes a softmax policy; for other policy classes, you need a different derivative. Second, the correction has a closed-form expression specifically because the slate is sampled *without replacement* from the softmax; if you relax this (e.g., sample with placement constraints), the correction becomes intractable.

4. Geyik, Ambler, Kenthapadi — *Fairness-Aware Ranking in Search & Recommendation Systems with Application to LinkedIn Talent Search* (KDD 2019)

The most ATS-relevant published fairness paper. LinkedIn's production recruiter-search architecture with a fairness post-processor enforcing gender-representation constraints on the shortlist.

Why it matters. The architecture — ML ranker + fairness LP post-processor — is exactly what you should implement for synthetic ATS. The paper documents extensive ablations on how to define the "qualified applicant pool" (the constraint reference distribution), which is the hardest research question in the area.

> "Our fairness-aware ranking algorithm enforces representation constraints at every rank cutoff. We achieve a 10–15% increase in the female representation of LinkedIn recruiter search results with negligible NDCG impact."

Commentary. The 10–15% delta is striking because the counterfactual is "do nothing". Most fairness interventions trade off significant accuracy for marginal fairness gains; this one gets a major fairness win at little accuracy cost. The reason: the production model was already close-to-fair on relevance, so the post-processor only needed to make small adjustments. If your baseline is more biased, the NDCG cost is higher.

5. Kull, Filho, Flach — *Beta Calibration: a well-founded and easily implemented improvement on logistic calibration for binary classifiers* (AISTATS 2017)

The three-parameter Beta-calibration family that beats Platt scaling on most binary classification tasks. The mechanism: $f_{\text{cal}}(s) = \sigma(a \log s + b \log(1-s) + c)$ with positive coefficients.

Why it matters. Beta is the right default calibrator in 2024+ for binary classifiers (hire / no-hire, bind / no-bind). It is two-line implementation in scikit-learn; it strictly dominates Platt; it does not need the $\sim$ 1000 samples that isotonic does.

> "We propose Beta calibration, a calibration method based on the Beta distribution that strictly generalises the logistic calibration of Platt (1999) while remaining a three-parameter family."

Commentary. The proof that Beta strictly generalises Platt is elegant: Platt is the special case $a = b$. Freeing $a \neq b$ allows the calibrator to handle skewed score distributions where the over-confidence is asymmetric across positive and negative classes. The catch: if your score is very confidently wrong on one tail, Beta still struggles — isotonic is the safer choice in the high-data regime.

Bonus paper: Liu, Mania, Jordan — *Competing Bandits in Matching Markets* (AISTATS 2020, arXiv:1906.05363)

The theoretical foundation for two-sided bandit matching. Both sides explore; both sides have preferences; the stable-matching regret bound is $\tilde{O}(K\sqrt{T})$ with constants dominated by the worst-served side.

Why it matters. The ATS market is two-sided. A unilateral CTR maximiser does not just lose welfare; it actively undermines its own training signal because the candidate side rejects the matches it suggests. This paper formalises that intuition.

> "We define the notion of *stable regret*, the difference between the expected match value of an algorithm and the worst stable match. Both UCB and TS-style algorithms achieve $\tilde{O}(K\sqrt{T})$ stable regret, matching the lower bound up to logarithmic factors."

Commentary. Stable regret is the right currency for two-sided markets. Pure-CTR regret is satisfied by an unstable matching (the recruiter accepts, the candidate withdraws); stable regret penalises that.

Bonus paper: Singh & Joachims — *Fairness of Exposure in Rankings* (KDD 2018, arXiv:1802.07281)

Introduces the exposure-fairness primitive: items get exposure equal to the position bias of the ranks they occupy; group fairness can be formalised as equal-exposure-per-group.

Why it matters. Exposure is the right unit of fairness for two-sided markets because attention is the scarce resource. Both synthetic insurance (brokers competing for lead-time-on-screen) and ATS (candidates competing for recruiter attention) are exposure-fairness problems fundamentally.

> "We propose a fairness constraint based on exposure, defined as the expected position-bias-weighted appearance of an item in the ranking. This formulation enables linear-program-based fair rerankers with provable accuracy guarantees."

Commentary. The LP formulation in the paper is solvable for top-50-sized slates in $< 10\,\text{ms}$; production deployable. The trickier question is what reference distribution to constrain against (parity? proportional? merit-weighted?) — the paper leaves this as an application choice.

Annotated reading order

Three weeks of evening reading:

Week 1: Joachims et al. WSDM 2017 (IPS-LTR), Ie et al. IJCAI 2019 (SlateQ), Chen et al. WSDM 2019 (REINFORCE-Top-K).

Week 2: Geyik et al. KDD 2019 (LinkedIn fairness), Singh & Joachims KDD 2018 (exposure), Kull et al. AISTATS 2017 (Beta calibration).

Week 3: Liu, Mania, Jordan AISTATS 2020 (competing bandits), Bottou et al. JMLR 2013 (counterfactual systems), Lattimore & Szepesvári ch. 32 (OPE) and ch. 21 (LinTS), Roth & Sotomayor *Two-Sided Matching* (1990, foundational).

Production papers (skim these too)

- Borisyuk, Kenthapadi, Salim — *LiJAR: LinkedIn Job Activity Reranker* (KDD 2017).
- Chandrashekar et al. — *Artwork Personalisation at Netflix* (Netflix blog, 2017).
- Steck — *Calibrated Recommendations* (RecSys 2018).
- McInerney, Lacker, Hansen et al. — *Explore, Exploit, and Explain: Personalising Spotify with Slate-Based RL* (RecSys 2018).
- Indeed Engineering Blog — *Job Search Two-Tower Retrieval* (2020–2022, multiple posts).
- ZipRecruiter Engineering Blog — *Two-Sided Marketplace Matching* (2021–2023).

Glossary of week 12 terms

Bi-encoder / two-tower. Two neural networks with a shared output dimension, scored by cosine similarity. Trained with InfoNCE on (query, positive-item) pairs with in-batch negatives.

Cross-encoder. A single neural network that takes the concatenated (query, item) sequence and outputs a relevance scalar. More accurate than a bi-encoder but lacks pre-indexing.

NDCG@k. Normalised Discounted Cumulative Gain at position k; the canonical graded-relevance ranking metric.

Position-bias. The probability that a user examines (and so can click) the item at position k; modelled as $p_k = 1/k^\eta$ in the canonical PBM.

IPS-LTR. Inverse-Propensity Scoring for Learning-to-Rank; weights each click by $1/p_k$ to remove position bias.

SlateQ. Slate Q-function decomposition under the single-choice user model; reduces slate optimisation to fractional knapsack.

Top-K correction. The $\lambda_K = K(1 - \pi(a|s))^{K-1}$ factor in slate-of-K policy gradients (Chen et al. WSDM 2019).

Demographic parity (DP). Equal probability of top-k inclusion across protected groups.

Equalised opportunity (EO). Equal probability of top-k inclusion among the *qualified* candidates across protected groups.

Exposure fairness. Equal position-bias-weighted exposure across protected groups (Singh & Joachims 2018).

ECE. Expected Calibration Error; bin-averaged absolute gap between predicted probability and empirical accuracy.

Beta calibration. $\sigma(a \log s - b \log(1 - s) + c)$ with $a, b \geq 0$; the recommended default calibrator for binary tasks.

Stable matching. A matching with no blocking pair (c, j) where each prefers the other to their current match.

FairCo. Online linear-program post-processor enforcing exposure-fairness constraints (Morik et al. SIGIR 2020).

Sourced from `research/bandits_matching_sota.md` sections 4–12 (CLTR, slate RL, matching markets, fairness, calibration, production).